Empowering Movement: A Human-in-the-Loop Proportional Controller for Assisted Elbow Flexion in Brachial Plexus Injury

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ABSTRACT

Introduction:

Although powered orthotic devices are becoming more available and increasingly recognized by the medical community, several factors still prevent them from being widely adopted. These orthoses typically use a simple controller, often termed a "bang-bang" controller, which actuates a motor if the muscle's electromyography (EMG) signal is above a set threshold. A proportional controller that could actuate the elbow mechanism based on the EMG signal magnitude would enable a better human-machine connection. The real challenge for designing a proportional controller lies in the heterogeneity of the types of injury and surgeries used in this patient population, as well as the extent of recovery post-surgery. Neural networks effectively model nonlinear relationships in control system design. Recently, a novel powered myoelectric elbow orthosis (PMEO) was developed and tested on patients with a brachial plexus injury (BPI) that used a bang-bang controller. The report evaluated the possibility of using a proportional controller using a neural network-powered regression (NNR) for this exoskeleton.

Materials and Methods:

Data from 31 participants with a BPI were collected after receiving IRB approval. A custom apparatus was designed to measure elbow flexion torque and EMG, while participants were instructed to match their torque to a predefined target (10%-40% of MVC). A 2-layered NNR (2 hidden nodes) with a Tanh activation and 5 cross-validation folds was used to find a relationship between the properties of the EMG (moving average, root mean square, variance, standard deviation, and slope) and the elbow torque for individual participants. The collected data were divided into a training set (70%) and a test set (30%). The root mean squared error (RMSE) for the test set was calculated for each NNR model, along with the training time and prediction speed.

Results:

A standard computer (Intel Core i5-10500 with 16 GB RAM) was used to train the models. The trained models required 82 seconds (range: 3-332 seconds) to train on 8.2e4 (range: 7.7e4-8.9e4) samples. The test dataset consisted of 3.5e4 (range: 3.3e4-3.8e4) samples. The median test RMSE was 0.39 Nm (range: 0.1-3.09 Nm). The NNR took 7 milliseconds (range: 6-10 milliseconds) to predict the test dataset; 35% and 16% of the models had an $R^2 > 0.5$ and >0.7, respectively.

Conclusions:

A low-powered office computer was sufficient to train an NNR with training time short enough to be completed in an orthotist's office within a typical appointment time. These data verify that the process can be field deployable.

INTRODUCTION

Peripheral nerve injuries (PNI) of the upper extremity are a significant consequence of combat trauma. During Operation Iraqi Freedom, PNIs accounted for approximately 6.0% of all battlefield injuries, with amputations comprising only 2.7% of upper extremity injuries, underscoring the high prevalence of nerve damage relative to limb loss.[1] Among these injuries, BPI, a particularly devastating form of PNI, frequently led to severely impaired upper limb function, which severely limits activities of daily living (ADLs) and results in long-term disability.[2,3] A BPI is nerve damage resulting from high-impact trauma. It disrupts the brachial plexus, impairing both movement and sensation in the upper arm and hand.[4,5] Surgical interventions, such as nerve grafts, nerve transfers, or free functioning muscle transfers (FFMT), were used to partially restore function but often failed to provide sufficient strength/control for independent limb use.[4,5] Combat-related PNIs and BPIs present unique challenges. The extent of neural injury often outweighs the availability of viable donor nerves for reconstruction, and as a result, complete functional restoration is extremely rare.[6] The donor nerve options available to reinnervate the elbow flexor also add complexity and variability of recovery.

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Powered orthotic devices are emerging as promising solutions to augment residual neuromuscular capacity. One such device is the novel powered myoelectric elbow orthosis (PMEO) developed at Mayo Clinic.[7,8] The PMEO was shown to improve the performance of the injured arm during ADL.[7] The PMEO control system relies on a simple on–off (bang-bang) style of torque control, which activates the motor when the electromyography (EMG) signal from the user's elbow flexor surpasses a set threshold.[8] Although effective in restoring some functional ability, such a control system does not capture the graded nature of human muscle activation, resulting in limited intuitiveness and increased user fatigue.

A proportional controller that maps EMG signal amplitudes to torque output can provide a more natural control. However, designing such controllers for patients with BPI is challenging because of heterogeneity in injury types, surgical reconstructions, and individual recovery trajectories.[9-11] Recent advances in neural networks have shown promise for modeling nonlinear input-output relationships in rehabilitation robotics and human-machine interfaces. George et al.[12] used a convolutional neural network (CNN) to create a human-in-the-loop model that predicted joint torque from EMG under different levels of exoskeleton assistance. Yang et al.[13] compared the accuracy of several models for pose estimation using surface EMG data. Kapelner et al.[14] used neural network regression (NNR) to non-linearly map motor neuron discharge timings to joint kinematics. These neural network models are computationally intensive and therefore not practical for training and deployment on resource-constrained or embedded devices in field settings. Achieving a balance between predictive accuracy and clinical usability remains a significant challenge in EMG-to-torque modeling. Methods requiring extensive calibration or high-performance computing hardware are often impractical for real-world clinical adoption. In this study, we evaluated the feasibility of implementing a 2-layer NNR-based proportional controller for a PMEO in the clinical setting. Specifically, we tested whether EMG features could reliably predict elbow torque across a cohort of individuals with BPI and assessed computational feasibility for clinic use to facilitate field deployment.

METHODS

Participants

A convenience sampling technique was used to recruit patients with a BPI because of the rare nature of the injury. Participants were eligible for participation in the study if data collection occurred at least 6 months following surgery, they were between 18 and 65 years of age and demonstrated a manual muscle testing (MMT) grade of 2 or greater. Exclusion criteria included having undergone a surgical procedure other than a nerve transfer, nerve graft, or FFMT, presenting with an MMT grade of M5 (as this would indicate a normal elbow flexor), or demonstrating refusal or inability to participate because of the difficulty following spoken instructions. Written informed consent was obtained as per Mayo Clinic's Institutional Review Board (IRB) guidelines (IRB Study nos. 19-008401 and 20-006259).

Experimental Setup

A custom apparatus was created to measure elbow flexion torque and surface EMG simultaneously (Figure 1A).[15,16] The participant's forearm was secured to the custom apparatus in 90° of elbow flexion. The torque produced by the elbow flexor was sensed by a torque sensor (TS11-20, Interface Inc.). A surface EMG sensor (MA100, Motion Lab Systems) was placed at the muscle belly of the elbow flexor in the direction of the muscle fibers (bicep brachii post nerve transfer and nerve graft or gracilis post FFMT). General procedures involved in EMG data collection were observed (i.e., cleaning the sensor, prepping the skin surface using alcohol pads, and securing the sensor to the skin while reducing any movement artifact). Participants were instructed to produce torque levels ranging from 10% to 40% of their maximal voluntary contraction (MVC) while receiving visual feedback of the torque target compared to their elbow flexor's torque (Figure 1B). The participants were not asked to output more than 40% of the MVC to reduce the amount of fatigue they experienced.[17] All data collection was performed on LabView 2019 (National Instruments, Austin, TX).

Feature Extraction

All the data were sampled at the rate of 500 Hz. The surface EMG data were filtered using a fourth order IIR bandpass Butterworth filter (15-200 Hz) to reduce background noise (as per the International Society of Electrophysiology and Kinesiology's recommendation[18]). The trained model would be deployed on an embedded system (within the PMEO) in the future. Hence, 5 time-domain features, which are computationally less demanding, were computed from a moving window of 100 samples of the EMG data:

1. Moving average (EMG_1): envelope of muscle activation
2. Root mean square (RMS) (EMG_2): envelope of muscle activation and muscle fatigue over sustained contraction
3. Variance (EMG_3): variability of muscle activation and fatigue over sustained contraction
4. Standard deviation (EMG_4): variability of muscle activation and action potential of motor units
5. Slope (EMG_5): Change in muscle activity

A min-max scaling technique was used to normalize the features and the torque (τ) between 0 and 1. Each feature vector corresponded to a specific elbow flexor torque value, that is, $\left[EMG_1, EMG_2, EMG_3, EMG_4, EMG_5\right]_i \triangleq \tau_i$, where i is the individual window.

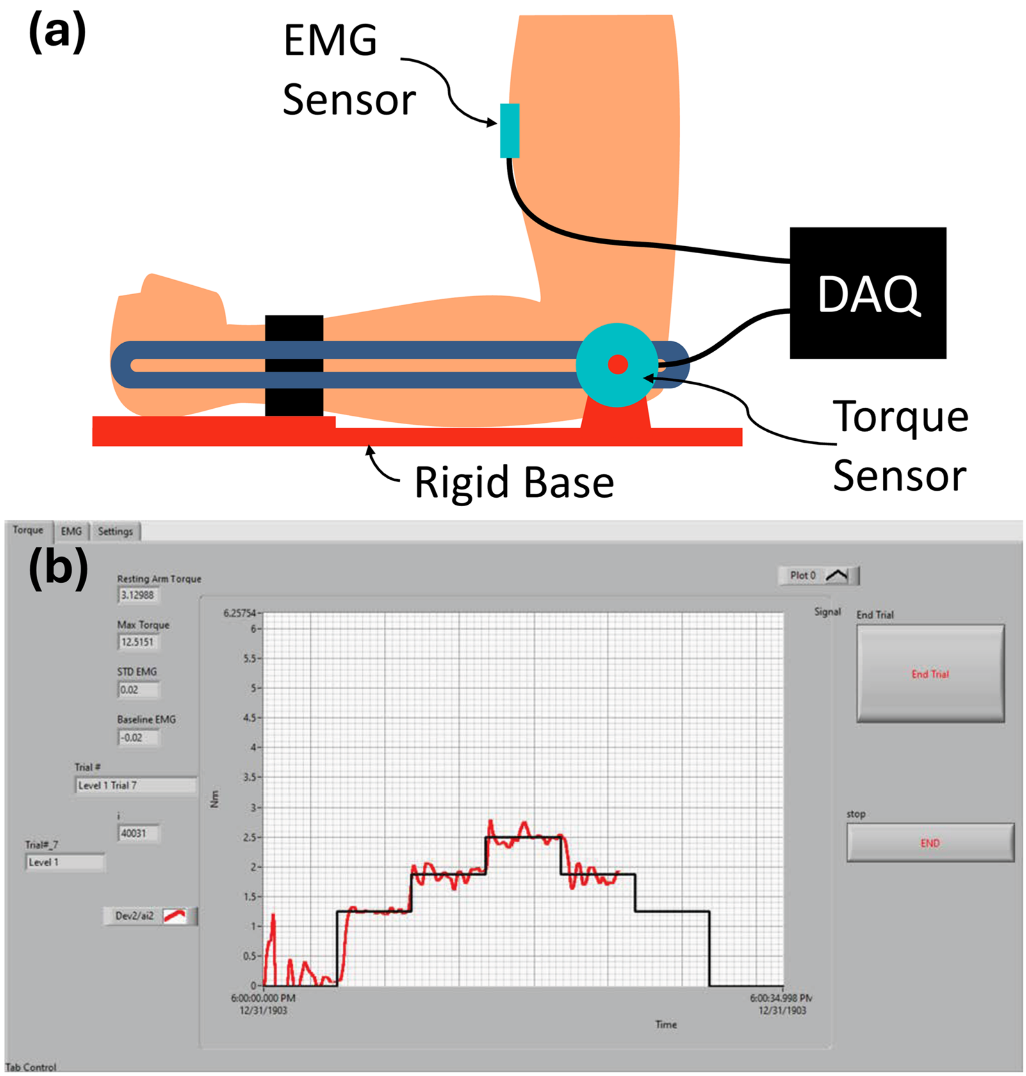

Figure 1. (A) Simplified experimental setup used to measure elbow flexion torque and the flexor muscle activity (EMG) at the elbow, (B) visual feedback displaying the elbow flexors' torque (red) and the target torques (black) on a custom LabView program.[16]

Neural Network Regression

A 2-layer neural network regression (NNR) model with Tanh activation and 2 hidden nodes was implemented because of its computational efficiency and feasibility for embedded deployment (Figure 2). A limited-memory Broyden-Fletcher-Goldfarb-Shanno quasi-Newton algorithm (LBFGS) was used as the loss function. To develop and evaluate the controller, the dataset was divided into training and testing subsets. For every torque measurement, 5 associated features ($EMG_{1\text{ to }5}$) were extracted and grouped as feature–torque pairs. These pairs were then randomly assigned to either the training or testing subset to ensure a representative distribution of torque values across both sets. A 70:30 split ratio was used, with 70% of the data allocated for training the model and 30% reserved for independent testing.[19] A separate NNR model was trained for individual participants. Training employed 5-fold cross-validation to minimize overfitting. The models were tested using only their respective participant's testing data set.

The performance of the model was determined using both its accuracy and the feasibility of the model in-the-field. Model accuracy was assessed using root mean squared error (RMSE) and coefficient of determination (R^2). Computational feasibility was evaluated based on training duration and prediction speed. All Models were trained on a standard desktop computer (Intel Core i5-10500, 16 GB RAM) using MATLAB 2021b (The MathWorks, Inc., Natick, Massachusetts, United States).

RESULTS

Participants

Thirty-one individuals (2 female/29 male) with chronic BPI participated in the study (Supplementary Material 1). The participants had a median age of 34 [interquartile range (IQR): 14; range: 19-65] years and were a median of 17 (IQR: 15; range: 7-155) months from the surgery. The study population included a diverse range of MMT grades, injury types, and nerve transfer procedures.

Neural Network Performance

The NNR models were trained on approximately 8.2e4 samples per subject (range: 7.7e4-8.9e4) (Figure 3A). Median RMSE across participants was 16% (range: 7%-28%) (Figure 3B). Training time was a median of 82 seconds (range: 3-332 seconds) (Figure 3C).

The testing dataset consisted of approximately 3.5e4 samples per subject (Figure 4A). Across participants, the models achieved a median RMSE of 16% of maximum voluntary contraction torque (range: 7%-8%) (Figure 4B). Prediction was computationally efficient, requiring a median of 7 milliseconds per dataset (range: 6-11 milliseconds), demonstrating suitability for near real-time implementation (Figure 4C). In terms of model fit, 35% of the trained models achieved an R^2 greater than 0.5, while 16% exceeded an R^2 of 0.7 (Figure 4D).

The best representative model fit was observed in one participant whose NNR achieved an R^2 value of 0.86 (Supplementary Material 2). The prediction speed for this participant's model was 8 milliseconds while testing.

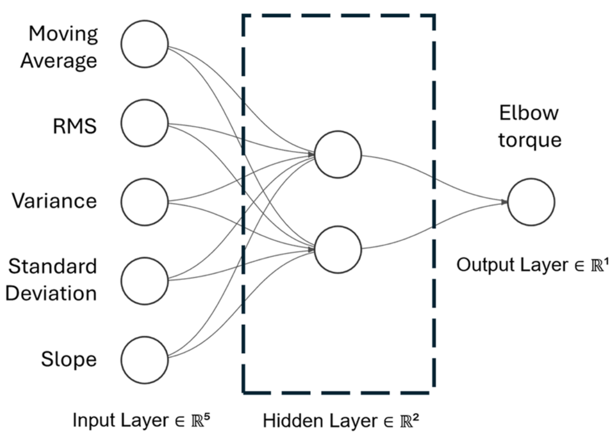

Figure 2. Neural network architecture. The input layer consisted of different features of the electromyography signals from the elbow flexor, with the output layer consisting of the elbow flexor torque.

DISCUSSION

The present study demonstrated that NNR can effectively model the nonlinear relationship between surface EMG features and elbow flexor torque in individuals with BPI. Training of the NNR models was computationally efficient, requiring a median of 82 seconds on a standard desktop computer, despite being trained on approximately 8.2e4 samples per subject. Comparatively, Wang and Buchanan[20] neural network model took 5 hours to be trained, although the computer they used to train the model had less power. This suggested that such models can be developed within clinically feasible time frames, such as during a standard orthotist appointment, without the need for high-performance computing resources.

Proportional Controllers are Better than Bang-Bang Controllers

Bang-bang controllers switch assistance on or off when EMG exceeds a threshold. Although this approach is simple and computationally inexpensive, it generally does not scale assistance with the magnitude of muscle activation, leading to abrupt torque delivery or a mismatch between device output and user effort.[21] Such sudden changes can be uncomfortable, destabilizing, or fatiguing, especially for rehabilitation users with compromised neuromuscular control. In contrast, proportional controllers are designed to scale assistance smoothly with EMG amplitude, making

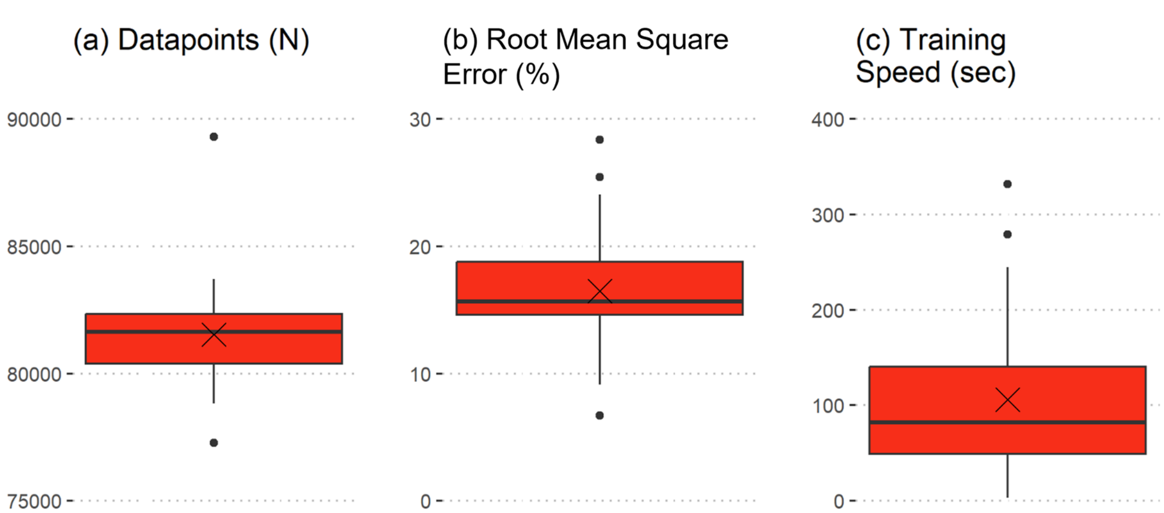

Figure 3. Performance parameters while training the neural networks. (a) Number of Datapoints, (b) Root Mean Square Error in percentage and (c) Training Speed in seconds. (X denotes the mean with outliers denoted by dots).

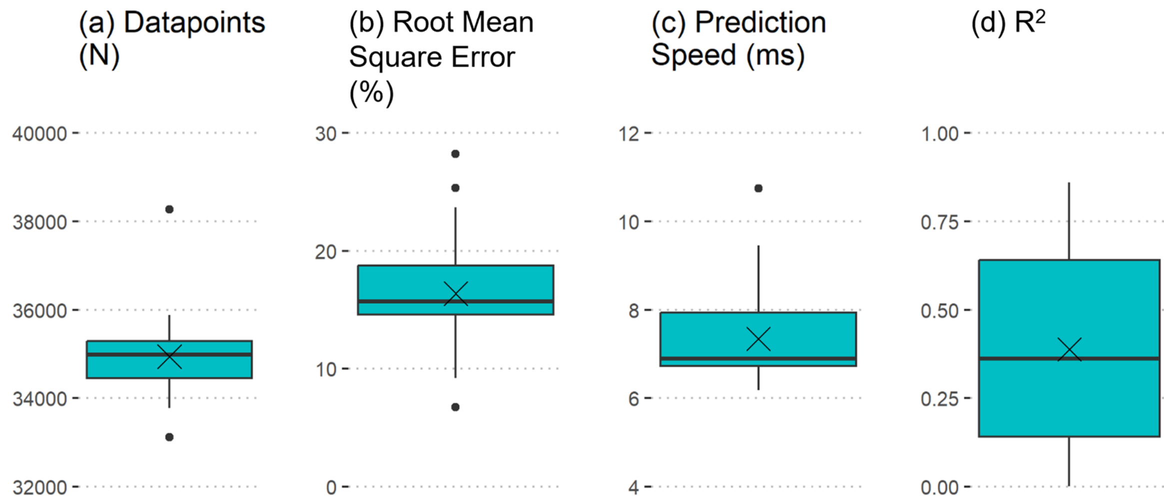

Figure 4. Performance parameters while testing the neural networks. (a) Number of Datapoints, (b) Root Mean Square Error in percentage, (c) Prediction Speed in milliseconds, and (d) Coefficient of Determination (R squared) (X denotes the mean with outliers denoted by dots).

them more physiologically intuitive. This aligns with the natural recruitment of motor units, where force generation increases proportionally with neural drive, thereby enhancing the intuitiveness of control. Claman reported that motor unit recruitment provides roughly linear force increase at lower demands, but at higher demands, the relationship is exponential.[22] In their review on the behavior of human motor units, Duchateau et al.[23] suggested that motor unit recruitment depends on the contraction speed and demand. This indicates the need of a more sophisticated neural network powered non-linear controller. Notably, in assistive devices, even simple proportional controllers have been reported to "allow users to perform fine or gross movements with ease".[24]

From a usability perspective, proportional control offers smoother transitions that reduce overshoot and jerky movements.[25] This improves comfort, safety, and stability during use, while also minimizing the likelihood that the user must "fight" against the device to correct for abrupt assistance. Such graded torque output is particularly valuable in tasks that require both precision and strength, such as holding lightweight objects versus lifting heavier loads, where bang-bang control may fail to adapt.[26] Over time, this smoother torque delivery may also reduce fatigue, since users are not forced to continuously counteract unintended or excessive assistance. Thus, the current results add to a growing body of literature suggesting that nonlinear proportional controllers represent a more intuitive, comfortable, and physiologically congruent approach for rehabilitation robotics and human–machine interaction.

Comparison with the NN Models in Literature

The NNR models in the present study achieved a median test RMSE of 16% MVC, which is higher than some high-precision models reported in the literature. For instance, Koirala et al.[27] and Dai et al.[28] demonstrated error rates as low as 5% MVC in able-bodied participants. Similarly, deep learning architectures such as convolutional and recurrent neural networks have yielded coefficients of determination (R^2) as high as 0.91 in healthy individuals.[29] However, it is important to emphasize that such high accuracies are often obtained under controlled laboratory conditions with consistent EMG patterns, extensive training datasets, and able-bodied participants. In contrast, the clinical context of the current study presents far greater challenges, including reduced and variable muscle activation, scarring, altered motor unit recruitment, and signal instability.[30,31] These factors inevitably degrade model performance compared to healthy cohorts, making a 16% MVC error highly acceptable and useful in this patient population.

High-precision deep learning models typically require substantial training time, large datasets, and computationally powerful hardware, which are not practical in the clinical workflow of orthotists or rehabilitation specialists. In this study, NNR training could be completed in a median of 82 seconds on a standard desktop computer, supporting the possibility of point-of-care deployment during a typical clinical appointment. This efficiency represents a critical balance between accuracy and usability, since methods that demand hours of calibration or specialized computing hardware are unlikely to be adopted in real-world clinical practice. Saad et al. stated that "a key factor to consider [while selecting models] is the interpretability or explainability of the model, which holds great importance, especially in healthcare applications."[32] The use of a relatively simple 2-layer NNR offers interpretability and transparency compared to deeper black-box models. Although many published models emphasize cross-subject generalization, this study focused on training participant-specific models. This approach will more realistically accommodate inter-subject variability that occurs in injured populations, especially injuries as complex as BPI.

Deployment to the Field/Orthotist Clinics

The observed training time (median 82 seconds) indicates that NNR models can be trained within the duration of a typical clinical appointment, making them practical for deployment in orthotist offices. This rapid training will allow clinicians to integrate the model directly into patient assessments, thereby supporting individualized device tuning without causing delays in workflow. Additionally, prediction times on the order of milliseconds enable real-time implementation in wearable systems, providing immediate feedback and adaptive control during patient movement. This capability has the potential to enhance patient safety, comfort, and mobility by allowing dynamic adjustment of orthoses during therapy sessions or daily activities.

The computational efficiency of NNR models further supports deployment on standard clinic hardware, including tablets, laptops, or embedded wearable controllers, hence minimizing the need for specialized computing infrastructure. Recent advancements in machine learning and AI have paved the way for such computational efficiency. Lattanzi et al.[33] developed an artificial neural network that could be run on wearable devices. To our knowledge, such a model with low memory requirements has not been developed and implemented on commercial exoskeletons.

Limitation

Only 35% of the NNR models in the current study achieved $R^2 > 0.5$, and only 16% exceeded $R^2 > 0.7$. Many participants had higher errors. This suggested a need for subject-specific calibration, or enrichment of data input. Although our simple architecture is beneficial for embedded/low-power deployment, more complex architectures might achieve lower errors. All of these could potentially improve the error rate but will also make the model more computationally expensive. Additionally, although the model was developed as a controller for the PMEO, it was never tested on the orthosis, with the current study only focusing on offline training. Future work includes deploying the model in a real-world simulation at an orthotist's office on patients with a BPI to test its real-time performance. Other potential future work is to analyze the NNR model's performance to MMT grades, injury types and time since surgery.

CONCLUSION

This study shows that NNR-based proportional controls for elbow flexion in patients with BPI are viable. The training time, prediction latency, and error levels are of the right order of magnitude to facilitate real-world deployment.

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SUPPLEMENTARY MATERIAL

Supplementary material is available at *Military Medicine* online.

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CONFLICT OF INTEREST STATEMENT

The authors have no competing interest to disclose.

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author, KRK, upon reasonable request.

INSTITUTIONAL REVIEW BOARD

The authors followed the guidelines set by Mayo Clinic's Institutional Review Board (IRB no. 19-008401 and 20-006259).

INSTITUTIONAL ANIMAL CARE AND USE COMMITTEE

Not applicable.

INSTITUTIONAL CLEARANCE

Institutional clearance does not apply.

INDIVIDUAL AUTHOR CONTRIBUTION STATEMENT

S.G.B. collected and analyzed the data. S.G.B. and K.R.K. designed the research study. A.Y.S. provided the clinical and military expertise related to brachial plexus injury. All authors were involved in the writing and review of the final manuscript.

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